

Population Exposure versus Personal Exposure. Population (or ambient) exposure refers to the average air pollution level measured in a grid cell. In contrast, personal exposure keeps track over the course of a day the exposure individuals encounter in different micro-environments, such as the freeway, outdoors and indoors. BenMAP-CE only represents population exposure.

Population-weighted Air Quality. Modeled or monitored ambient concentrations that have been weighted according to the number of people exposed.

Prevalence Rate. The percentage of individuals in a given population who already have a given adverse health condition. Used to calculate changes in health conditions among those who already have a health condition, such as asthmatics.

Random Effects Pooling. Random effect pooling is one method to combine two or more distributions of health impact or economic value estimates into a single new distribution. This approach allows the possibility that the estimated parameter from different studies may in fact be estimates of different parameters, rather than just different estimates of a single underlying parameter.

Regulatory Impact Analysis (RIA). A policy tool used to assess the likely effects of a proposed regulation or regulatory change. It usually involves detailed analyses to quantify the costs and benefits of the regulation.

Relative Risk. Relative risk typically is used as a measure of the change in risk of an adverse health effect associated with an increase in air pollution levels in an epidemiology study. More specifically, it is the ratio of the risk of illness with a higher pollution level to the risk of illness with a lower pollution level, where the “risk” is defined as the probability that an individual will become ill.

Rollback. The process by which monitor data are reduced to a different level. BenMAP-CE rolls back monitor data in three ways. Percentage rollback reduces all monitor observations by the same percentage. Incremental rollback reduces all observations by the same increment. Rollback to a standard reduces monitor observations so that they just meet a specified standard.

Setup. A BenMAP-CE setup encapsulates all of the data needed to run analyses for a particular geographic area—a city, an entire country, etc. These data consist of grid definitions, pollutants, monitor data, incidence and prevalence rates, population data, health impact functions, variables, inflation rates, and valuation functions.

Shapefile. A shapefile is a particular type of GIS file, and has a *.shp* extension. These files are accompanied by companion files with *.shx* and *.dbf* extensions, and can be used to create Shapefile Grid Definitions. See <http://www.esri.com/library/whitepapers/pdfs/shapefile.pdf> for more information.

Sum Dependent Pooling. Summing two or more incidence or valuation results, assuming the underlying functions are correlated. For example, summing the incidence

of respiratory hospital admissions for two different age groups quantified using C-R functions from the same study.

Sum Independent Pooling. Summing two or more incidence or valuation results, assuming the underlying functions are independent (uncorrelated). For example, summing the incidence of respiratory hospital admissions quantified using C-R functions from different studies using different methods.

Threshold. BenMAP-CE's advanced settings for health impact functions allows you to specify an air quality threshold; this is an air quality level below which benefits are not calculated. For example, if the threshold is 5 $\mu\text{g}/\text{m}^3$, then only areas with PM_{2.5} concentrations equal to or greater than 5 $\mu\text{g}/\text{m}^3$ will be included in estimating health incidence results. Specifying a threshold does not affect the shape of the C-R function used to quantify impacts.

Unit Value. A unit value is the estimated mean economic value of avoiding a single case of a particular health effect.

User-defined Weights Pooling. User-defined weights let you specify the weights that you want to use when combining two or more distributions of results. The weights should sum to one. If not, BenMAP-CE normalizes the weights so that they do.

Valuation Function. Valuation functions are used by BenMAP-CE to estimate the economic values of changes in the incidence of health effects. These are selected within an Aggregation, Pooling, and Valuation Configuration (APV Configuration).

Variable Datasets. Health Impact functions and valuation functions may sometimes refer to variables other than those for which BenMAP-CE automatically calculates values. For example, some valuation functions reference the wage data within each area of analysis. To facilitate this, BenMAP-CE allows you to load datasets of variables for use in functions, which may be used globally or may vary geographically (meaning they are associated with a particular Grid Definition).

VNA (Voronoi Neighbor Averaging). An algorithm used by BenMAP-CE to interpolate air quality monitoring data to an unmonitored location. BenMAP-CE first identifies the set of monitors that best “surround” the center of the population grid cell and then takes an inverse-distance weighted average of the monitoring values. This is discussed in detail in Appendix B.

WTP (Willingness to Pay). The willingness of individuals to pay for a good or service, such as a reduction in the risk of illness. In general, economists tend to view an individual's WTP for an improvement in environmental quality as the appropriate measure of the value of a risk reduction. An individual's willingness to accept (WTA) compensation for not receiving an improvement is also a valid measure. However, WTP is generally considered to be a more readily available and conservative measure of benefits.

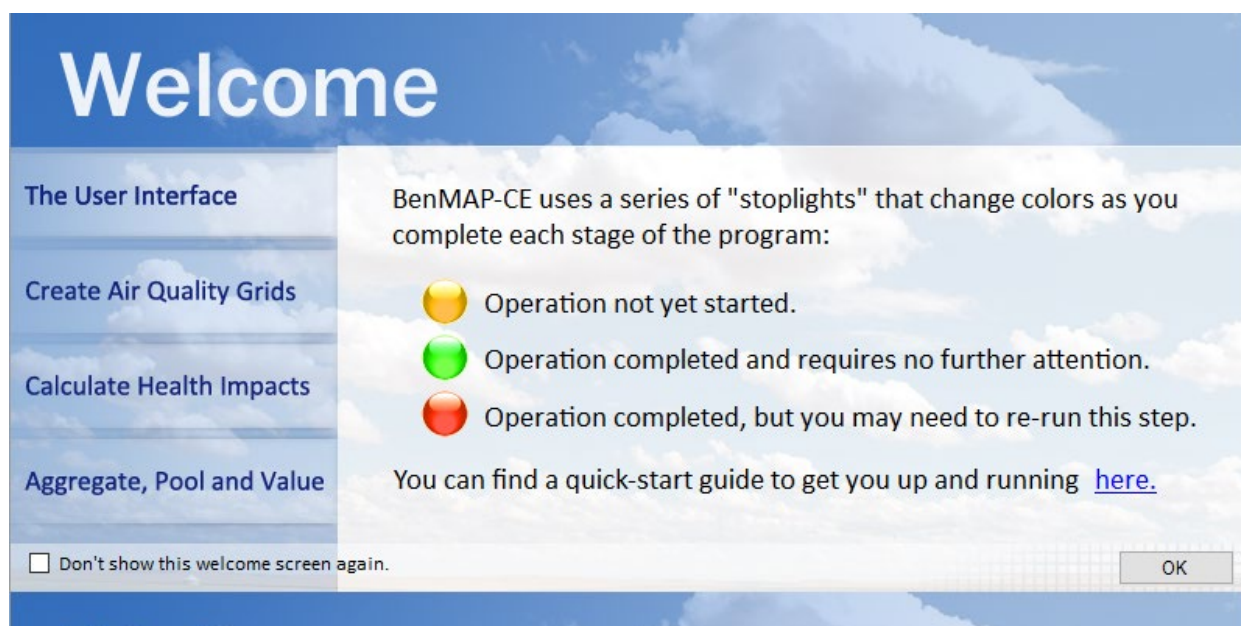
Chapter 3

Overview of BenMAP-CE Features

In this chapter...

- **Get an overview of the features available with the Core Program.**
- **Learn about additional BenMAP-CE modules.**
- **Learn about the Tools and Help menu options.**
- **Find descriptions of the various outputs including types of files, results, maps and reports available from BenMAP-CE.**

Upon starting BenMAP-CE for the first time, you will see the following **Welcome** screen.



The **Welcome** screen gives a brief description of the user interface and highlights the “stoplight” metaphor used in BenMAP-CE to indicate the status of analytical steps performed using the tree menu on the left side of the main window. Clicking the links on the left side of the **Welcome** screen (e.g., **Create Air Quality Grids**) will provide information about each feature. You will also find a link to the BenMAP-CE website for downloading the most current BenMAP-CE software and other reference information. If you do not wish to see the **Welcome** screen at program start-up, check the option ‘*Don't show this window screen again*’ in the lower left-hand corner of the screen. To re-enable the **Welcome** screen, go to **Tools** menu on the main BenMAP-CE window, select **Options**, and check the option for *Show Start Window*. Press **OK** on the **Welcome** screen to close this window and display the BenMAP-CE main window.